Test Series Code

04

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 18/01/2026

ANSWER KEY

1.	(2)	46.	(1)	91.	(4)	136.	(3)
2.	(1)	47.	(1)	92.	(3)	137.	(4)
3.	(1)	48.	(4)	93.	(1)	138.	(4)
4.	(1)	49.	(1)	94.	(3)	139.	(2)
5.	(2)	50.	(4)	95.	(1)	140.	(2)
6.	(3)	51.	(4)	96.	(4)	141.	(3)
7.	(1)	52.	(3)	97.	(4)	142.	(2)
8.	(2)	53.	(1)	98.	(1)	143.	(2)
9.	(2)	54.	(4)	99.	(4)	144.	(3)
10.	(3)	55.	(2)	100.	(3)	145.	(2)
11.	(3)	56.	(4)	101.	(2)	146.	(1)
12.	(2)	57.	(4)	102.	(3)	147.	(2)
13.	(1)	58.	(1)	103.	(1)	148.	(2)
14.	(4)	59.	(1)	104.	(1)	149.	(1)
15.	(2)	60.	(1)	105.	(3)	150.	(1)
16.	(3)	61.	(3)	106.	(4)	151.	(4)
17.	(2)	62.	(4)	107.	(3)	152.	(2)
18.	(2)	63.	(2)	108.	(3)	153.	(1)
19.	(2)	64.	(2)	109.	(3)	154.	(3)
20.	(2)	65.	(4)	110.	(1)	155.	(4)
21.	(3)	66.	(3)	111.	(2)	156.	(2)
22.	(3)	67.	(2)	112.	(4)	157.	(3)
23.	(1)	68.	(2)	113.	(1)	158.	(2)
24.	(3)	69.	(3)	114.	(3)	159.	(1)
25.	(2)	70.	(4)	115.	(1)	160.	(1)
26.	(3)	71.	(1)	116.	(2)	161.	(2)
27.	(3)	72.	(2)	117.	(4)	162.	(1)
28.	(2)	73.	(4)	118.	(3)	163.	(3)
29.	(1)	74.	(3)	119.	(4)	164.	(4)
30.	(4)	75.	(4)	120.	(3)	165.	(2)
31.	(4)	76.	(1)	121.	(1)	166.	(4)
32.	(1)	77.	(1)	122.	(3)	167.	(4)
33.	(3)	78.	(3)	123.	(1)	168.	(4)
34.	(2)	79.	(2)	124.	(2)	169.	(2)
35.	(3)	80.	(4)	125.	(3)	170.	(3)
36.	(1)	81.	(4)	126.	(4)	171.	(1)
37.	(2)	82.	(1)	127.	(3)	172.	(3)
38.	(2)	83.	(2)	128.	(4)	173.	(1)
39.	(1)	84.	(3)	129.	(2)	174.	(2)
40.	(1)	85.	(3)	130.	(2)	175.	(2)
41.	(1)	86.	(1)	131.	(3)	176.	(4)
42.	(3)	87.	(4)	132.	(2)	177.	(2)
43.	(2)	88.	(1)	133.	(4)	178.	(2)
44.	(3)	89.	(4)	134.	(1)	179.	(3)
45.	(4)	90.	(2)	135.	(4)	180.	(1)

[Contd...

Hints & Solutions

1. (2)

Using First Law: $dQ = dU + dW$

$CdT = C_v dT + dW$

$\left(C_v + \frac{RT}{T_0}\right)dT = C_v dT + dW \Rightarrow dW = \frac{RT}{T_0}dT$

$W = \int_{T_0}^{2T_0} \frac{R}{T_0} T dT = \frac{R}{T_0}[\frac{T^2}{2}]_{T_0}^{2T_0}$

$W = \frac{R}{2T_0}\left[4T_0^2 - T_0^2\right] = 1.5RT_0$.

(NCERT Ed. 2025-26, 11th, Page No. 231)

2. (1)

According to Kepler's Second Law of planetary motion, a satellite in an elliptical orbit moves fastest when it is closest to the central body (perigee) and slowest when it is farthest away (apogee).

As the satellite moves away from Earth (towards apogee), it slows down.

As the satellite moves towards Earth (towards perigee), it speeds up.

Diagram (1) correctly illustrates this principle.

(NCERT Ed. 2025-26, 11th, Page No. 129)

3. (1)

According to the law of equipartition of energy, the average rotational kinetic energy for any diatomic molecule (like O_2 and N_2) is $2\times\left(\frac{1}{2}kT\right) = kT$ at the same temperature. It does not depend on the mass or the number of moles.

(NCERT Ed. 2025-26, 11th, Page No. 252)

4. (1)

Total expansion

$\Delta L = \Delta L_c + \Delta L_s = L_c\alpha_c\Delta T + L_s\alpha_s\Delta T$.

$\alpha_{eff} = \frac{\Delta L}{L_{total}\Delta T} = \frac{L_c\alpha_c + L_s\alpha_s}{L_c + L_s}$.

$\alpha_{eff} = \frac{(1\times1.7 + 2\times1.1)\times10^{-5}}{1+2} = \frac{1.7+2.2}{3}\times10^{-5}$

$= \frac{3.9}{3}\times10^{-5} = 1.3\times10^{-5}/^{o}\text{C}$

(NCERT Ed. 2025-26, 11th, Page No. 205)

5. (2)

Spinning ball and non spinning balls have different trajectory. Pressure is lower on the side of the ball where the velocity of air relative to the ball is higher.

(NCERT Ed. 2025-26, 11th, Page No. 189)

6. (3)

$T^2 \propto R^3$

$\frac{T_1}{T_2} = \left(\frac{R_1}{R_2}\right)^{3/2}$

$\frac{200}{T_2} = \left(\frac{1}{\frac{1}{4}}\right)^{3/2}$

T_2 = 25 days

Time taken to complete 4 revolution

= 25 × 4 = 100 days

(NCERT Ed. 2025-26, 11th, Page No. 129)

7. (1)

Bernoulli's equation: $P + \frac{1}{2}\rho v^2 = \frac{P}{2} + \frac{1}{2}\rho(2v)^2$

$\frac{P}{2} = \frac{1}{2}\rho\left(4v^2 - v^2\right) = \frac{3}{2}\rho v^2$

$\Rightarrow P = 3\rho v^2$

(NCERT Ed. 2025-26, 11th, Page No. 188)

8. (2)

Rate of cooling

$\frac{dT}{dt} = \frac{\sigma AeT^4}{ms} = \frac{\sigma\left(4\pi R^2\right)eT^4}{\left(\frac{4}{3}\pi R^3\rho\right)s} \propto \frac{R^2}{R^3} \propto \frac{1}{R}$

(NCERT Ed. 2025-26, 11th, Page No. 220)

9. (2)

$f_{\max} = \mu \times N = 0.5\times40 = 20$ N

Downward forces:

$30\text{ N} + \text{Weight}\left(2\text{ kg}\times10\right) = 50\text{ N}$

[Contd...

Upward force: 15 N

$F_{\text{net}} = 50\text{ N} - 15\text{ N} = 35\text{ N}$ (downward)

Since the downward force (35 N) is greater than the maximum possible friction (20 N), the block slides down. Therefore, friction acts at its maximum value in the opposite direction.

Friction = 20 N , Upward.

(NCERT Ed. 2025-26, 11th, Page No. 61)

10. (3)

Area under graph = $\frac{1}{2}\times(P_0 + 2P_0)\times V_0$

$= \frac{3}{2}P_0V_0$

(NCERT Ed. 2025-26, 11th, Page No. 235)

11. (3)

$v_t \propto r^2$

If $r \to 2r$, $v_{t'} \propto (2r)^2 = 4r^2$

$v_{t'} = 4v$.

(NCERT Ed. 2025-26, 11th, Page No. 192)

12. (2)

Work-Energy Theorem ($W = \Delta KE = \int Pdt$).

$\int_0^2 \frac{3t^2}{2}dt = \left[\frac{3t^3}{2\times 3}\right]_0^2 = \frac{2^3}{2} = 4$ J.

$\Delta KE = \frac{1}{2}mv_f^2 - 0 \Rightarrow 4 = \frac{1}{2}\times 2\times v_f^2$.

$v_f^2 = 4 \Rightarrow v_f = 2$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 83)

13. (1)

Heat lost by Copper

$= ms\Delta T = 2000\text{ g}\times 0.1\times 100 = 20000$ cal

Heat gained by Ice $= m_{ice}L = m_{ice}\times 80$

$80m_{ice} = 20000 \Rightarrow m_{ice} = 250$ g

(NCERT Ed. 2025-26, 11th, Page No. 212)

14. (4)

Using the result = $V = \sqrt{\frac{2gH}{1+\frac{H}{R}}}$

$b^2 v_e^2 = \frac{2gH}{1+\frac{H}{R}} \qquad \left(v_e = \sqrt{2gr}\right)$

$\Rightarrow \quad b^2(2gR) = \frac{2gH}{1+\frac{H}{R}}$

$\Rightarrow \quad RH = (R+H)\left(b^2R\right)$

$H = (R+H)b^2$

$\Rightarrow \quad H - b^2H = b^2R$

$H = \frac{b^2R}{1-b^2}$

(NCERT Ed. 2025-26, 11th, Page No. 135)

15. (2)

Velocity of efflux $v = \sqrt{2g(H-h)}$

Time to fall $t = \sqrt{2h/g}$

Range

$R = v\times t = \sqrt{2g(H-h)}\sqrt{\frac{2h}{g}} = 2\sqrt{h(H-h)}$

For max R , $\frac{dR}{dh} = 0 \Rightarrow h = H - h \Rightarrow h = \frac{H}{2}$

(NCERT Ed. 2025-26, 11th, Page No. 189)

16. (3)

$f_{mix} = \frac{n_1f_1 + n_2f_2}{n_1+n_2} = \frac{1(3)+1(5)}{1+1} = \frac{8}{2} = 4$

(NCERT Ed. 2025-26, 11th, Page No. 253)

17. (2)

Wien's Law: $\lambda_m T = \text{constant}$.

If $T \to 2T$, $\lambda_m \to \frac{\lambda_m}{2} = \frac{500}{2} = 250$ nm .

(NCERT Ed. 2025-26, 11th, Page No. 218)

18. (2)

$\eta = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{600} = 1 - 0.5 = 0.5$

or 50%

(NCERT Ed. 2025-26, 11th, Page No. 238)

[Contd...

19. (2)

Statement I is incorrect: Plane angle and solid angle are dimensionless but have units (Radian and Steradian).

Statement II is correct: If a quantity has no unit, it cannot have dimensions (as dimensions are powers to which fundamental units are raised).

(NCERT Ed. 2025-26, 11th, Page No. 2)

20. (2)

$P_{bottom} = P_{atm} + \rho gh = 3P_{atm}$

$\rho gh = 2P_{atm} = 2\times10^5$

$1000\times10\times h = 2\times10^5$

$\Rightarrow 10^4 h = 2\times10^5 \Rightarrow h = 20$ m

(NCERT Ed. 2025-26, 11th, Page No. 183)

21. (3)

Because the horizontal speed magnitude is constant, the total horizontal distance traveled by the ball (sum of all segments) must be equal to the theoretical range R.

Total horizontal distance = $d + d + x$

$R = 2d + x$

$x = R - 2d$

(NCERT Ed. 2025-26, 11th, Page No. 40)

22. (3)

$v_{rms} = \sqrt{3RT/M}$.

If $T \to 4T$ and $M \to M/2$ (dissociation),

$v' = \sqrt{3R(4T)/(M/2)} = \sqrt{8\cdot(3RT/M)}$

$= \sqrt{8}v = 2.82v$

(NCERT Ed. 2025-26, 11th, Page No. 250)

23. (1)

$P = \frac{1}{3}\frac{N}{V}mv_{rms}^2$

Initial Pressure: $P = \frac{1}{3}\frac{N}{V}mv_{rms}^2$

Final Mass: $m' = 2m$

Final RMS Speed: $v'_{rms} = \frac{v_{rms}}{2}$

$P' = \frac{1}{3}\frac{N}{V}(2m)\left(\frac{v_{rms}}{2}\right)^2$

$P' = \frac{1}{3}\frac{N}{V}(2m)\left(\frac{v_{rms}^2}{4}\right)$

$P' = \frac{1}{2}\left[\frac{1}{3}\frac{N}{V}mv_{rms}^2\right]$

$P' = \frac{P}{2}$

(NCERT Ed. 2025-26, 11th, Page No. 249)

24. (3)

An air bubble inside a liquid has only one free surface (the interface where air meets water).

$P_1 = \frac{2T_{water}}{R}$

so $P_1 = \frac{2T}{R}$

A soap bubble in air has two free surfaces (an inner surface and an outer surface, both in contact with air).

$P_2 = \frac{4T_{soap}}{R}$

$P_2 = \frac{4(0.5T)}{R} = \frac{2T}{R}$

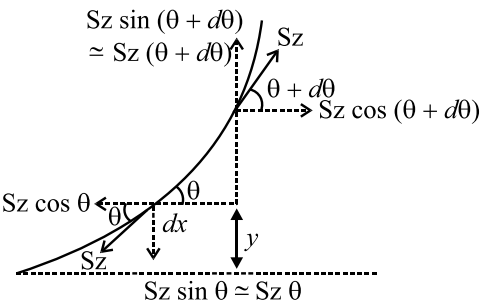
Ratio: Comparing the two: $\frac{P_1}{P_2} = \frac{\frac{2T}{R}}{\frac{2T}{R}} = 1$

The ratio is 1:1.

(NCERT Ed. 2025-26, 11th, Page No. 196)

25. (2)

Consider a small element of the surface of width dx. The vertical component of surface tension force at any point x is $F_y = Sz\sin\theta \approx Sz\theta$.



The net upward force on the element due to the difference in angles at x and $x + dx$ is:

[Contd...

$dF_{tension} = Sz(\theta + d\theta) - Sz\theta = Szd\theta$

This force balances the weight of the liquid column of height y (relative to the flat surface at $x = 0$):

$dF_{gravity} = (\rho \cdot y \cdot dx \cdot z)g$

$Szd\theta = \rho y dx z g$

$\frac{d\theta}{dx} = \frac{\rho g y}{S}$

(NCERT Ed. 2025-26, 11th, Page No. 195)

26. (3)

$\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$

Where:

d is the diameter of the molecule.

n is the number density (number of molecules per unit volume, $n = N/V$).

Since the mass density ρ is given by

$\rho = \frac{M_{total}}{V} = \frac{N \cdot m}{V} = n \cdot m$

(where m is the mass of one molecule), it follows that: $\rho \propto n$

When the gas is compressed isothermally such that the mass density ρ is doubled, the number density n also doubles.

From the formula,

$\lambda \propto \frac{1}{n}$

Therefore, if n becomes 2*n*, the new mean free path λ' becomes: $\lambda' = \frac{\lambda}{2}$

(NCERT Ed. 2025-26, 11th, Page No. 255)

27. (3)

Newton's Law of Cooling:

$\frac{T_1 - T_2}{t} = K\left(\frac{T_1 + T_2}{2} - T_s\right)$

Case 1: $\frac{80-50}{5} = K\left(\frac{130}{2} - 20\right)$

$\Rightarrow 6 = 45K \Rightarrow K = \frac{6}{45} = \frac{2}{15}$

Case 2: $\frac{60-30}{t} = \frac{2}{15}\left(\frac{90}{2} - 20\right)$

$\Rightarrow \frac{30}{t} = \frac{2}{15}(25) = \frac{50}{15} = \frac{10}{3}$

$t = \frac{90}{10} = 9\text{min}$

(NCERT Ed. 2025-26, 11th, Page No. 221)

28. (2)

Centripetal Force: $F_c = \frac{mv^2}{r}$

Hypothetical Gravitational Force: $F_g = \frac{k}{r^3}$

$\frac{mv^2}{r} = \frac{k}{r^3}$

$v^2 = \frac{k}{mr^2}$

$v \propto \frac{1}{r}$

$T = \frac{2\pi r}{v}$

Substitute the proportionality for *v*:

$T \propto \frac{r}{\frac{1}{r}}$

$T \propto r^2$

(NCERT Ed. 2025-26, 11th, Page No. 129)

29. (1)

Volume remains constant:

$2 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \Rightarrow R = 2^{1/3} r$

Initial Surface Energy $U_i = 2 \times \left(4\pi r^2 T\right) = 8\pi r^2 T$

Final Surface Energy

$U_f = 4\pi R^2 T = 4\pi (2^{1/3} r)^2 T = 4\pi r^2 \cdot 2^{2/3} T$

Ratio $\frac{U_i}{U_f} = \frac{8}{4 \cdot 2^{2/3}} = \frac{2}{2^{2/3}} = 2^{1/3} : 1$

(NCERT Ed. 2025-26, 11th, Page No. 195)

30. (4)

Stefan-Boltzmann Law: $E \propto T^4$.

If $T \to T/2$, then $E' \propto (T/2)^4 = T^4/16$.

$E' = \frac{E}{16}$

(NCERT Ed. 2025-26, 11th, Page No. 219)

[Contd...

31. (4)

In the steady state of a uniform rod, the emperature gradient $\left(\frac{\Delta T}{\Delta x}\right)$ is constant. This means the temperature decreases linearly with distance.

Temperature drop over 40 cm

$= 100°C - 70°C = 30°C$

Temperature gradient $\left(\frac{\Delta T}{\Delta x}\right) = \frac{30°C}{40\text{ cm}} = 0.75°C/cm$

Total temperature drop

= gradient $\times$ $L = 0.75 \times 100 = 75°C$.

Temperature at the other end (T_2) = (Initial Temperature) – (Total Drop)

$T_2 = 100°C - 75°C = 25°C$

(NCERT Ed. 2025-26, 11th, Page No. 215)

32. (1)

Elastic Potential Energy

$U = \frac{1}{2} \times \text{Stress} \times \text{Strain} \times \text{Volume}$ or $U = \frac{F^2 L}{2AY}$.

$F = mg = 10 \times 10 = 100$ N.

$0.1 = \frac{100^2 \times 2}{2 \times 10^{-6} \times Y}$.

$0.1 = \frac{10000}{10^{-6} Y} \Rightarrow Y = \frac{10^4}{0.1 \times 10^{-6}} = \frac{10^4}{10^{-7}} = 10^{11}$ N/m^2

(NCERT Ed. 2025-26, 11th, Page No. 174)

33. (3)

$V_{in} = -\frac{GM}{2R^3}(3R^2 - r^2)$

Since the potential $\frac{4}{3}V(1.33V)$ lies between the surface potential (V) and the center potential ($1.5V$), the point must be inside the sphere.

$V_{in} = \frac{4}{3}V : -\frac{GM}{2R^3}(3R^2 - r^2) = \frac{4}{3}\left(-\frac{GM}{R}\right)$

$\frac{1}{2R^2}(3R^2 - r^2) = \frac{4}{3}$

$3(3R^2 - r^2) = 8R^2$

$9R^2 - 3r^2 = 8R^2$

$R^2 = 3r^2 \Rightarrow r = \frac{R}{\sqrt{3}}$

(NCERT Ed. 2025-26, 11th, Page No. 134)

34. (2)

For isobaric process: $W = P\Delta V = nR\Delta T$

$Q = nC_p\Delta T = n\left(\frac{\gamma R}{\gamma - 1}\right)\Delta T = \frac{\gamma}{\gamma - 1}W$

$Q = \frac{1.4}{1.4 - 1} \times 200 = \frac{1.4}{0.4} \times 200 = 3.5 \times 200 = 700$ J

(NCERT Ed. 2025-26, 11th, Page No. 253)

35. (3)

Height of liquid level will be maximum in vessel *C* hence the pressure as well as force at the base due to liquid will be maximum in vessel *C*.

(NCERT Ed. 2025-26, 11th, Page No. 183)

36. (1)

Thermal Stress $= Y\alpha\Delta T = \left(2 \times 10^{11}\right) \times \left(10^{-5}\right) \times 10$

$= 2 \times 10^7$ N/m^2

(NCERT Ed. 2025-26, 11th, Page No. 207)

37. (2)

$W = 2 \times 4\pi T\left(r_2^2 - r_1^2\right)$ (2 surfaces for soap bubble).

$W = 8\pi T\left(2^2 - 1^2\right) = 8\pi T(3) = 24\pi T$.

$W' = 8\pi T\left(3^2 - 2^2\right) = 8\pi T(5) = 40\pi T$.

$W' = \frac{40}{24}W = \frac{5}{3}W \approx 1.66W$

(NCERT Ed. 2025-26, 11th, Page No. 194)

38. (2)

Conservation of Angular Momentum ($L_i = L_f$).

Initial Moment of Inertia $I_i = \frac{ML^2}{12}$ (beads at center have $r = 0$).

Final Moment of Inertia

$I_f = \frac{ML^2}{12} + 2 \times m(L/2)^2 = \frac{ML^2}{12} + \frac{mL^2}{2}$

$= \frac{ML^2 + 6mL^2}{12}$

$I_i\omega_0 = I_f\omega_f \Rightarrow \left(\frac{ML^2}{12}\right)\omega_0 = \left(\frac{M + 6m}{12}\right)L^2\omega_f$

$\omega_f = \frac{M\omega_0}{M + 6m}$.

(NCERT Ed. 2025-26, 11th, Page No. 122)

[Contd...

39. (1)

$$K.E. = \frac{GMm}{2(R+h)}$$

$$P.E. = -\frac{GMm}{R+h} = -2(K.E.)$$

$$T.E. = K.E. + P.E. = \frac{-GMm}{2(R+h)}$$

(NCERT Ed. 2025-26, 11th, Page No. 138)

40. (1)

Let t_1 be acceleration time and t_2 be deceleration time. $T = t_1 + t_2$.

$V_{max} = \alpha t_1$ and $V_{max} = \beta t_2$.

$t_1 = \frac{V_{max}}{\alpha}$, $t_2 = \frac{V_{max}}{\beta}$.

$$T = V_{max}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) = V_{max}\left(\frac{\alpha+\beta}{\alpha\beta}\right).$$

$$V_{max} = \frac{\alpha\beta T}{\alpha+\beta}$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

41. (1)

$\Delta Q = -20$ J (released). $\Delta W = -8$ J (done on gas).

$\Delta U = \Delta Q - \Delta W = -20 - (-8) = -12$ J

$U_f - U_i = -12 \Rightarrow U_f = 30 - 12 = 18$ J

(NCERT Ed. 2025-26, 11th, Page No. 230)

42. (3)

$$h = \frac{2T\cos\theta}{r\rho g_{eff}}$$

In free fall, $g_{eff} = 0$. Mathematically $h \to \infty$.

The water will rise to the full available length of the tube but will *not* overflow (due to surface tension constraints at the edge).

(NCERT Ed. 2025-26, 11th, Page No. 197)

43. (2)

Pressure on the concave side is higher than the convex side.

(NCERT Ed. 2025-26, 11th, Page No. 196)

44. (3)

For an adiabatic process, the relationship between pressure and volume is:

$PV^{\gamma} = \text{constant}$

$\Rightarrow P_1V_1^{\gamma} = P_2V_2^{\gamma}$

Given:

Initial Pressure $P_1 = P$

Initial Volume $V_1 = V$

Final Volume $V_2 = \frac{V}{8}$

For a monoatomic gas, $\gamma = \frac{5}{3}$

Rearranging to find P_2: $P_2 = P_1\left(\frac{V_1}{V_2}\right)^{\gamma}$

$$P_2 = P\left(\frac{V}{V/8}\right)^{5/3}$$

$P_2 = P(8)^{5/3}$

To solve $8^{5/3}$ easily:

$8^{5/3} = (8^{1/3})^5$

Since $8^{1/3} = 2$:

$P_2 = P(2)^5 = 32P$

(NCERT Ed. 2025-26, 11th, Page No. 235)

45. (4)

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$v_{avg} = \sqrt{\frac{8RT}{\pi M}}$$

$$v_{mp} = \sqrt{\frac{2RT}{M}}$$

$$v_{rms} : v_{mp} : v_{avg} = \sqrt{3} : \sqrt{2} : \sqrt{\frac{8}{\pi}}$$

(NCERT Ed. 2025-26, 11th, Page No. 256)

[Contd...

46. (1)

- $\underset{\text{Base}}{NH_2^-} \xrightarrow{+H^+} \underset{\text{Conjugate acid}}{NH_3}$
- Electron deficient species like $AlCl_3$ can act as Lewis acid.

(NCERT Ed. 2025-26 11th Part-I Page No. 191)

47. (1)

- At anode:

$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$

- At cathode:

$2H_2O(\ell) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$

(NCERT Ed. 2025-26 12th Part-I Page No. 53)

48. (4)

$$M = \frac{\%\text{ by Mass} \times d(g/ml) \times 10}{\text{Molar mass of solute}}$$

$$= \frac{2 \times 1.5 \times 10}{36.5}$$

$$= 0.82 \text{ mol } L^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 23)

49. (1)

$$AgCl \rightleftharpoons \underset{S}{Ag^+} + \underset{S}{Cl^-}$$

$$CaCl_2 \rightarrow \underset{0.1\,M}{Ca^{2+}} + \underset{0.2\,M}{2Cl^-}$$

$$K_{sp} = [Ag^+][Cl^-]$$

$$1.8 \times 10^{-10} = [Ag^+][0.2]$$

$$[Ag^+] = 9 \times 10^{-10} M$$

(NCERT Ed. 2025-26 11th Part-I Page No. 206)

50. (4)

$$\underset{1\,mol}{I_2} \longrightarrow 2I^{+5} + \underset{10\,F}{10e^-}$$

Charge required = 10 × 0.1 = 1 F

(NCERT Ed. 2025-26 12th Part-I Page No. 52)

51. (4)

$$MnO_4^- \xrightarrow[\text{n-factor} = 5]{H^+} Mn^{2+}$$

$$I^- \xrightarrow{\text{n-factor} = 1} \frac{1}{2}I_2 + 1e^-$$

eq. of $KMnO_4$ = eq. of I^-

$$n \times 5 = 1 \times 1$$

$$n = \frac{1}{5}$$

(NCERT Ed. 2025-26 11th Part-II Page No. 246)

52. (3)

$$Cl_2(g) \rightleftharpoons 2Cl(g)$$

$$K_p = \frac{(p_{Cl})^2}{(p_{Cl_2})}$$

$$= \frac{(1 \times 0.4)^2}{1 \times 0.6} = 0.26$$

(NCERT Ed. 2025-26 11th Part-I Page No. 183)

53. (1)

- For weak electrolytes solution, specific conductance decreases on dilution.
- The electronic conductance decreases with increase of temperature.

(NCERT Ed. 2025-26 12th Part-I Page No. 42, 49)

54. (4)

$Na(Z = 11): 1s^2 2s^2 2p^6 3s^1$

Last electron of Na atom is present in 3s orbital; so for that electron

$$n = 3, \ell = 0, m = 0, s = \pm\frac{1}{2}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 63)

55. (2)

On increasing the volume; pressure will decrease and reaction will move in that direction where number of gaseous moles increases.

(NCERT Ed. 2025-26 11th Part-I Page No. 186)

56. (4)

On charging the battery $PbSO_4(s)$ on cathode is converted into $PbO_2(s)$.

(NCERT Ed. 2025-26 12th Part-I Page No. 55)

57. (4)

Mixture of chloroform and acetone shows negative deviation from Raoult's law.

So, $\Delta H < 0, \Delta V < 0, \Delta G < 0$ but $\Delta S > 0$.

(NCERT Ed. 2025-26 12th Part-I Page No. 13)

[Contd...

58. (1)

- $K_c = \frac{1}{[Fe^{3+}][OH^-]^3} M^{-4}$
- $K_c = [NO_2]^4[O_2] M^5$
- $K_c = [CO_2]\ M$
- $K_c = \frac{[I_2][Cl_2]}{[ICl]^2}$

(NCERT Ed. 2025-26 11th Part-I Page No. 180)

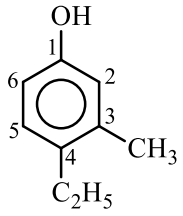
59. (1)

$\Lambda_m^o(CH_3COOH) = \Lambda_m^o(HNO_3) + \Lambda_m^o(CH_3COONa) - \Lambda_m^o(NaNO_3)$

$= x_2 + x_3 - x_1\ S\,cm^2\ mol^{-1}$

(NCERT Ed. 2025-26 12th Part-I Page No. 50)

60. (1)



4-Ethyl-3-methylphenol

(NCERT Ed. 2025-26 11th Part-II Page No. 269)

61. (3)

$M_3V_3 = M_1V_1 + M_2V_2$

$M_3 \times 300 = 0.1 \times 100 + 0.2 \times 200$

$M_3 = \frac{50}{300} = \frac{1}{6}$

$pH = -\log\left(\frac{1}{6}\right) = \log 2 + \log 3$

$= 0.3010 + 0.4771 = 0.778$

(NCERT Ed. 2025-26 11th Part-I Page No. 194)

62. (4)

Lesser the reduction potential, more will be reducing power.

(NCERT Ed. 2025-26 12th Part-I Page No. 37)

63. (2)

$\underset{\substack{1\,mol\\(58\,g)}}{C_4H_{10}(g)} + \frac{13}{2}\underset{6.5\,mol}{O_2(g)} \rightarrow 4CO_2(g) + 5H_2O(\ell)$

Volume of O_2 required at STP

$= \frac{6.5 \times 22.4}{58} \times 5.8\ L$

$= 14.56\ L$

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

64. (2)

$K_1 = \frac{[C]}{[A][B]^2}$

$K_2 = \frac{[A]^2[B]^4}{[C]^2}$

$\therefore\ K_1^2K_2 = 1$

$K_1 = \frac{1}{\sqrt{K_2}}$

(NCERT Ed. 2025-26 11th Part-I Page No. 176)

65. (4)

$M^{n+} + ne^- \rightarrow M$

$W = ZIt = \frac{M}{\text{n-factor} \times 96500} \times i \times t$

$\therefore\ \text{mole deposit} = \frac{i \times t}{\text{n-factor} \times 96500}$

$2.5 \times 10^{-3} = \frac{1 \times 965}{n \times 96500}$

$n = \frac{1 \times 965}{2.5 \times 10^{-3} \times 96500}$

n = 4

(NCERT Ed. 2025-26 12th Part-I Page No. 51)

66. (3)

- $S_2F_2 : 2x + 2(-1) = 0$
  $\therefore\ x = +1$
- $SO_2 : x + 2(-2) = 0$
  $\therefore\ x = +4$
- $S_8 : 8x = 0$
  $\therefore\ x = 0$

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

67. (2)

$pH = \frac{1}{2}\left[p_{K_w} + pK_a + \log C\right]$

$= \frac{1}{2}\left[14 + 4.74 + \log(10^{-1})\right] = 8.87$

(NCERT Ed. 2025-26 11th Part-I Page No. 202)

[Contd...

68. (2)

$\Delta_r G^o = -nE^oF = -2 \times 1.05 \times 96500 \text{ J mol}^{-1}$

$= -202650 \text{ J mol}^{-1}$

$= -202.65 \text{ kJ mol}^{-1}$

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

69. (3)

The valence electronic configuration of Cr is $3d^5 4s^1$ because due to relatively small shielding, smaller coulombic repulsion energy and larger exchange energy, the electronic configuration $3d^5$ has extra stability.

(NCERT Ed. 2025-26 11th Part-I Page No. 64)

70. (4)

$CH_4(g) + 2S_2(g) \rightleftharpoons CS_2(g) + 2H_2S(g)$

$\Delta n_g = 3 - 3 = 0$

$K_p = K_c(RT)^0 = K_c$

(NCERT Ed. 2025-26 11th Part-I Page No. 178)

71. (1)

$2H^+ + 2e^- \rightarrow H_2$

$E_{H^+/H_2} = E^o_{H^+/H_2} - \frac{0.059}{2} \log \frac{1}{[H^+]^2}$

$-0.413 = 0 - \frac{0.059}{2} \log [H^+]^{-2}$

$0.413 = 0.059 \text{ pH} \quad \therefore \text{ pH} = 7$

(NCERT Ed. 2025-26 12th Part-I Page No. 36)

72. (2)

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(\ell)$

$\Delta H = \Sigma(\Delta H_f)_P - \Sigma(\Delta H_f)_R$

$= \left[(\Delta_f H)_{CO_2} + 2(\Delta_f H)_{H_2O}\right] - \left[(\Delta_f H)_{CH_4}\right]$

$= y + 2z - x \text{ kJ mol}^{-1}$

(NCERT Ed. 2025-26 11th Part-I Page No. 150)

73. (4)

$Zn(OH)_2(s) \rightleftharpoons Zn^{2+}(aq) + 2OH^-(aq)$

For $\text{pH} = 12, \text{pOH} = 14 - 12 = 2$

$\therefore [OH^-] = 10^{-2} \text{ M}$

$\therefore K_{sp} = (S)(10^{-2})^2$

$1 \times 10^{-15} = S \times 10^{-4}$

$S = 1 \times 10^{-11} \text{ M}$

(NCERT Ed. 2025-26 11th Part-I Page No. 206)

74. (3)

In the fuel cell energy of combustion of fuel is directly converted into electrical energy.

(NCERT Ed. 2025-26 12th Part-I Page No. 56)

75. (4)

- Reverse osmosis is used in desalination of sea water.
- Excess of salty food results a situation called edema.

(NCERT Ed. 2025-26 12th Part-I Page No. 23)

76. (1)

$NH_2COONH_4(s) \rightleftharpoons \underset{2p}{2NH_3(g)} + \underset{p}{CO_2(g)}$

Given: $0.9 = 2p + p = 3p$

$p = 0.3 \text{ atm}$

$K_p = (2p)^2(p) = 4p^3$

$= 4(0.3)^3 = 0.108 \text{ atm}^3$

(NCERT Ed. 2025-26 11th Part-I Page No. 180)

77. (1)

$\rho = \frac{RA}{L} = \frac{2.22 \times 10^3 \times 0.1}{111} = 2 \text{ ohm cm}$

$\kappa = \frac{1}{\rho} = \frac{1}{2} \text{S cm}^{-1}$

$\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{1}{2} \times \frac{1000}{0.01} = 5 \times 10^4 \text{ S cm}^2 \text{ mol}^{-1}$

(NCERT Ed. 2025-26 12th Part-I Page No. 45)

78. (3)

Law of multiple proportions in not applicable for the oxide(s) of magnesium because magnesium can form only one oxide (MgO).

(NCERT Ed. 2025-26 11th Part-I Page No. 15)

79. (2)

Common ion effect is observed for a mixture containing weak electrolyte (NH_4OH) and strong electrolyte (NH_4Cl), having a common ion (NH_4^+).

(NCERT Ed. 2025-26 11th Part-I Page No. 206)

[Contd...

80. (4)

Daniell cell: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, $E^{o}_{cell} = 1.1\,V$

When E_{ext} (opposite) $= E^{o}_{cell}$, no flow of current or electrons.

(NCERT Ed. 2025-26 12th Part-I Page No. 32)

81. (4)

Disproportionation reaction	$Cl_2 + 2NaOH \rightarrow NaClO + NaCl + H_2O$
Non-redox reaction	$CaCO_3 \rightarrow CaO + CO_2$
Non-metal displacement reaction	$Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$
Metal displacement reaction	$CuSO_4 + Zn \rightarrow Cu + ZnSO_4$

(NCERT Ed. 2025-26 11th Part-II Page No. 242)

82. (1)

$$2AB(g) \rightleftharpoons A_2(g) + B_2(g)$$

(*i*) 1 0 0

(f) $1-\alpha$ $\frac{\alpha}{2}$ $\frac{\alpha}{2}$

$$K_p = \frac{(p_{A_2})(p_{B_2})}{(p_{AB})^2} = \frac{\left(\frac{P\frac{\alpha}{2}}{1}\right)\left(\frac{P\frac{\alpha}{2}}{1}\right)}{\left(\frac{p(1-\alpha)}{1}\right)^2}$$

$$K_p = \frac{\alpha^2}{4(1-\alpha)^2} \simeq \frac{\alpha^2}{4}$$

$$\therefore\ \alpha = 2\sqrt{K_p}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 181)

83. (2)

$M^{4+} + e^- \rightarrow M^{3+}$, $E^{o}_1 = 2.4\,V$

$M^{3+} + e^- \rightarrow M^{2+}$, $E^{o}_2 = 0.6\,V$

$M^{4+} + 2e^- \rightarrow M^{2+}$, E^{o}_3

$$\Delta G^{o}_3 = \Delta G^{o}_1 + \Delta G^{o}_2$$

$$E_3 = \frac{n_1E^{o}_3 + n_2E^{o}_2}{n_3}$$

$$= \frac{1\times 2.4 + 1\times 0.6}{2} = 1.5\,V$$

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

84. (3)

On increasing the intensity of incident light during photoelectric effect, kinetic energy of emitted electrons remain same but number of photoelectrons increases.

$$KE = h\nu - W_0$$

(NCERT Ed. 2025-26 11th Part-I Page No. 41)

85. (3)

$$p_{OH} = 14 - 11 = 3$$

$$[OH^-] = 10^{-3}\,M$$

$$[OH^-] = C\alpha$$

$$10^{-3} = 0.1\times\alpha$$

$$\therefore\ \alpha = 10^{-2}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 197)

86. (1)

- E_{cell} is an intensive property.
- Electrochemical cells are extensively used for determining the equilibrium constant.

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

87. (4)

- Exothermic process may be spontaneous or non-spontaneous.
- A system at higher temperature has greater randomness in it than one at lower temperature.

(NCERT Ed. 2025-26 11th Part-I Page No. 159)

88. (1)

On adding SO_2Cl_2 both equilibria will affected and amount of SO_2 and $COCl_2$ will increase but amount of CO will decrease.

(NCERT Ed. 2025-26 11th Part-I Page No. 186)

89. (4)

$$\lambda^{o} : H^+ > Mg^{2+} > K^+ > Na^+$$

(NCERT Ed. 2025-26 12th Part-I Page No. 49)

90. (2)

For a galvanic cell:

$\Delta G_{cell} < 0$ and $E_{cell} > 0$

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

[Contd...

91. (4)

Tap roots of carrot, turnip and adventitious roots of sweet potato, get swollen and store food. The hanging structures that support a banyan tree are called prop roots. Similarly, the stems of maize and sugarcane have supporting roots coming out of the lower nodes of the stem. These are called stilt roots. In some plants such as *Rhizophora* growing in swampy areas, many roots come out of the ground and grow vertically upwards. Such roots, called pneumatophores, help to get oxygen for respiration.

(NCERT Ed. 2022-23, 11th, Page No. 67)

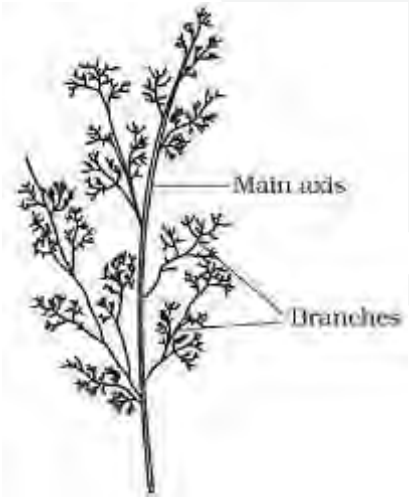
92. (3)

The Mycoplasma are organisms that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen. They are members of kingdom Monera.

(NCERT Ed. 2025-26, 11th, Page No. 14)

93. (1)

The alga given in the question is *Polysiphonia*, a red alga. The food is stored as floridean starch which is very similar to amylopectin and glycogen in structure.



Polysiphonia

(NCERT Ed. 2025-26, 11th, Page No. 27)

94. (3)

Biological names are generally in Latin and written in italics. They are Latinised or derived from Latin irrespective of their origin. The first word in a biological name represents the genus while the second component denotes the specific epithet. The specific epithet i.e. species represents the lowest category of the taxonomical hierarchy. Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their Latin origin. The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter. Name of the author appears after the specific epithet, i.e., at the end of the biological name and is written in an abbreviated form, e.g., *Mangifera indica* Linn. It indicates that this species was first described by Linnaeus.

(NCERT Ed. 2025-26, 11th, Page No. 4)

95. (1)

Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms whereas fertilisation restores the diploid phase. A bivalent consists of two homologous chromosomes, which in turn are each made up of two sister chromatids, totaling four chromatids. A diploid cell with '2n' chromosomes will form 'n' bivalents during meiosis I. Following the two meiotic divisions, four haploid gametes are produced, each containing 'n' chromosomes. Therefore, the number of bivalents formed during meiosis I is indeed equal to the number of chromosomes present in a gamete of that organism.

(NCERT Ed. 2025-26, 11th, Page No. 126)

96. (4)

Antonie Von Leeuwenhoek first saw and described a live cell. Robert Brown later discovered the nucleus. The invention of the microscope and its improvement leading to the electron microscope revealed all the structural details of the cell. In 1838, Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant. At about the same time, Theodore Schwann (1839), a German Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells. Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.

(NCERT Ed. 2025-26, 11th, Page No. 88)

[Contd...

97. (4)

In early days, human beings needed to find sources for their basic needs of food, clothing and shelter. Hence, the earliest classifications were based on the 'uses' of various organisms. The artificial system of classification separated the closely related species since they were based on a few characteristics. They were based mainly on vegetative characters or on the androecium structure (system given by Linnaeus). Also, the artificial systems gave equal weightage to vegetative and sexual characteristics; this is not acceptable since we know that often the vegetative characters are more easily affected by environment.

(NCERT Ed. 2025-26, 11th, Page No. 23)

98. (1)

The members of phaeophyceae or brown algae are found primarily in marine habitats. They possess chlorophyll *a*, *c*, carotenoids and xanthophylls. Food is stored as complex carbohydrates, which may be in the form of laminarin or mannitol. The vegetative cells have a cellulosic wall usually covered on the outside by a gelatinous coating of algin. The gametes are pyriform (pear-shaped) and bear two laterally attached flagella.

(NCERT Ed. 2025-26, 11th, Page No. 26)

99. (4)

In pea and bean flowers, there are five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel); this type of aestivation is known as vexillary or papilionaceous.

(NCERT Ed. 2025-26, 11th, Page No. 79)

100. (3)

The pteridophytes are further classified into four classes: Psilopsida (*Psilotum*); Lycopsida (*Selaginella*, *Lycopodium*), Sphenopsida (*Equisetum*) and Pteropsida (*Dryopteris*, *Pteris*, *Adiantum*).

(NCERT Ed. 2025-26, 11th, Page No. 32)

101. (2)

Synthesis of genetic material: This process, also known as DNA replication, occurs during the S phase of interphase.

Formation of chromosomal tetrads: This event takes place during Prophase I of meiosis I. Homologous chromosomes pair up in a process called synapsis to form structures called bivalents or tetrads.

Formation of daughter cells with half the number of chromosomes in each: This is the result of Meiosis I. At the end of Telophase I and cytokinesis, the original diploid cell divides into two haploid daughter cells. Each of these cells contains half the number of chromosomes as the parent cell, although each chromosome still consists of two sister chromatids.

Condensation of chromosomes after interkinesis: Interkinesis is a brief resting period between meiosis I and meiosis II. Following interkinesis, the cell enters Prophase II, where the chromosomes condense again in preparation for the second meiotic division.

Separation of sister chromatids: This event occurs during Anaphase II of meiosis II. The centromeres holding the sister chromatids together divide, and the sister chromatids are pulled apart to opposite poles of the cell. Each separated chromatid is now considered an individual chromosome.

(NCERT Ed. 2025-26, 11th, Page No. 126)

102. (3)

Societies which indulged in anthropocentric view of biology could register limited progress in biological knowledge. This implies a human-centered view, as the text contrasts it with the later, more objective systematic description of life forms. Systematic and monumental description of life forms brought in, out of necessity, detailed systems of identification, nomenclature and classification. The biggest spin off of such studies was the recognition of the sharing of similarities among living organisms both horizontally and vertically. That all present day living organisms are related to each other and also to all organisms that ever lived on this earth, was a revelation which humbled man and led to cultural movements for conservation of biodiversity.

(NCERT Ed. 2025-26, 11th, Page No. 1)

103. (1)

Puccinia is a basidiomycete. The asexual spores are generally not found, but vegetative reproduction by fragmentation is common. *Aspergillus* is an ascomycete. Sexual spores are called ascospores which are produced endogenously in sac like asci (singular ascus). *Albugo* is a phycomycete. The mycelium is aseptate and coenocytic. *Trichoderma* is a deuteromycete. Members of deuteromycetes are commonly known as imperfect fungi because only the asexual or vegetative phases of these fungi are known.

(NCERT Ed. 2025-26, 11th, Page No. 17)

104. (1)

In marginal placentation the placenta forms a ridge along the ventral suture of the ovary. When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentaion is said to be axile. In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part. When the ovules are borne on central axis and septa are absent, as in *Dianthus* and *Primrose* the placentation is called free central. In basal placentation, the placenta develops at the base of ovary.

(NCERT Ed. 2025-26, 11th, Page No. 65)

105. (3)

Malvaceae is commonly called the cotton family. The fruit in members of family Graminae is generally a caryopsis.

(NCERT Ed. 2025-26, 11th, Page No. 64)

106. (4)

In the seeds of cereals such as maize the seed coat is membranous and generally fused with the fruit wall. The endosperm is bulky and stores food. The outer covering of endosperm separates the embryo by a proteinous layer called aleurone layer.

(NCERT Ed. 2025-26, 11th, Page No. 67)

107. (3)

In a pinnately compound leaf a number of leaflets are present on a common axis, the rachis, which represents the midrib of the leaf as in neem. A bud is present in the axil of petiole in both simple and compound leaves, but not in the axil of leaflets of the compound leaf.

(NCERT Ed. 2025-26, 11th, Page No. 60)

108. (3)

The liverworts as well as mosses have foot, seta and capsule in their sporophytes. The leaves of *Pinus* are needle-like, which is an adaptation to reduce the surface area. In heterosporous pteridophytic genera like *Selaginella* and *Salvinia* the megaspores and microspores germinate and give rise to female and male gametophytes, respectively. The female gametophytes in these plants are retained on the parent sporophytes for variable periods. The development of the zygotes into young embryos take place within the female gametophytes. This event is a precursor to the seed habit considered an important step in evolution. *Eucalyptus* is an angiosperm.

(NCERT Ed. 2025-26, 11th, Page No. 32)

109. (3)

Prokaryotic cells: There is no well-defined nucleus. No organelles, like the ones in eukaryotes, are found in prokaryotic cells except for ribosomes. The eukaryotic ribosomes are 80S while the prokaryotic ribosomes are 70S. A specialised differentiated form of cell membrane called mesosome is the characteristic of prokaryotes. The genetic material is basically naked, not enveloped by a nuclear membrane.

(NCERT Ed. 2025-26, 11th, Page No. 90)

110. (1)

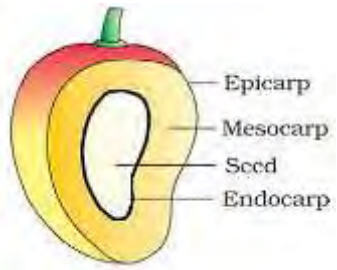
Ernst Mayr, the Harvard University evolutionary biologist who has been called 'The Darwin of the 20th century', was one of the 100 greatest scientists of all time. Mayr was awarded the three prizes widely regarded as the triple crown of biology: the Balzan Prize in 1983, the International Prize for Biology in 1994, and the Crafoord Prize in 1999.

(NCERT Ed. 2025-26, 11th, Page No. 2)

111. (2)

Generally, the fruit consists of a wall or pericarp and seeds. The pericarp may be dry or fleshy. When pericarp is thick and fleshy, it is differentiated into the outer epicarp, the middle mesocarp and the inner endocarp. In mango the pericarp is well differentiated into an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp.

[Contd...



Parts of a fruit: Mango

(NCERT Ed. 2025-26, 11th, Page No. 65)

112. (4)

Mycorrhiza is a symbiotic association between a fungus (a eukaryote) and the roots of a higher plant (also a eukaryote) e.g. *Pinus*. In a lichen, the fungal partner is the mycobiont and the algal partner is the phycobiont. Mycorrhiza is a symbiotic association, not a single organism, and therefore it is not classified in any kingdom. The two organisms involved belong to Kingdom Fungi and Kingdom Plantae.

(NCERT Ed. 2025-26, 11th, Page No. 32)

113. (1)

The red algae reproduce asexually by non-motile spores and sexually by non-motile gametes. Union of gametes in brown algae may take place in water or within the oogonium (oogamous species). Most green algae have one or more storage bodies called pyrenoids located in the chloroplasts. Some of the blue-green algae can fix atmospheric nitrogen in specialised cells called heterocysts.

(NCERT Ed. 2025-26, 11th, Page No. 27)

114. (3)

Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle. Cells in this stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.

(NCERT Ed. 2025-26, 11th, Page No. 122)

115. (1)

The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte. They lack true roots, stem or leaves. The male sex organ is called antheridium. Zygotes do not undergo reduction division immediately. They produce a multicellular body called a sporophyte. The sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it.

(NCERT Ed. 2025-26, 11th, Page No. 29)

116. (2)

In plants such as bean, gram and pea, the endosperm is not present in mature seeds and such seeds are called non-endospermous. The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.

(NCERT Ed. 2025-26, 11th, Page No. 66)

117. (4)

The characteristic inflorescence of the Compositae family is a capitulum or head, which is a type of racemose inflorescence. In the Compositae family, the gynoecium is bicarpellary and syncarpous, but the ovary is unilocular (one-chambered) with a single ovule and basal placentation. The ovary is inferior.

(NCERT Ed. 2025-26, 11th, Page No. 65)

118. (3)

Leguminosae - $\% \, ♂ K_{(5)} \, C_{1+2+(2)} \, A_{(9)+1} \, \underline{G}_1$

Solanaceae - $\oplus ♂ K_{(5)} \, \overbrace{C_{(5)}} \, A_5 \, \underline{G}_{(2)}$

Cruciferae (Brassicaceae) - $\oplus ♂ K_{2+2} \, C_4 \, A_{2+4} \, \underline{G}_{(2)}$

Graminae - Br (Lemma), Brl (Palea)

♀, ☿, P_{2-3} lodicules, A_3 or $_{3+3}$, G_1 or $G_{(3)}$

(NCERT Ed. 2025-26, 11th, Page No. 67)

119. (4)

In *Spirogyra*, the sexual reproduction occurs by means of non-flagellated and similar sized gametes.

(NCERT Ed. 2025-26, 11th, Page No. 24)

[Contd...

120. (3)

Stem tendrils which develop from axillary buds, are slender and spirally coiled and help plants to climb such as in gourds (cucumber, pumpkins, watermelon) and grapevines. Axillary buds of stems may also get modified into woody, straight and pointed thorns. Thorns are found in many plants such as *Citrus*, *Bougainvillea*. Some plants of arid regions modify their stems into flattened (*Opuntia*), or fleshy cylindrical *(Euphorbia)* structures. They contain chlorophyll and carry out photosynthesis. Underground stems of potato, ginger, turmeric, *zaminkand*, *Colocasia* are modified to store food in them. Underground stems of some plants such as grass and strawberry, etc., spread to new niches and when older parts die new plants are formed.

(NCERT Ed. 2022-23, 11th, Page No. 68)

121. (1)

Chemotaxonomy uses the chemical constituents of the plant to resolve confusions.

(NCERT Ed. 2025-26, 11th, Page No. 24)

122. (3)

Viruses did not find a place in classification since they are not considered truly 'living', if we understand living as those organisms that have a cell structure. The viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell. Viruses are obligate parasites.

In addition to proteins, viruses also contain genetic material, that could be either RNA or DNA.

(NCERT Ed. 2025-26, 11th, Page No. 19)

123. (1)

Animal cells have centrioles which are absent in almost all plant cells. Chromatin contains DNA and some basic proteins called histones, some non-histone proteins and also RNA. DNA is present in nucleus, mitochondria and chloroplasts. RNA is present in nucleus, cytoplasm, mitochondria and chloroplasts.

(NCERT Ed. 2025-26, 11th, Page No. 91)

124. (2)

In pteridophytes, the sporangia produce spores by meiosis in spore mother cells. The spores germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes (haploid) called prothallus.

(NCERT Ed. 2025-26, 11th, Page No. 32)

125. (3)

The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system. The smooth endoplasmic reticulum is the major site for synthesis of lipid. In animal cells lipid-like steroidal hormones are synthesised in SER. The golgi apparatus principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell. Golgi apparatus is the important site of formation of glycoproteins and glycolipids. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes (hydrolases – lipases, proteases, carbohydrases) optimally active at the acidic pH. These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids. Photosynthesis and ATP synthesis are functions of chloroplast and mitochondria, respectively.

(NCERT Ed. 2025-26, 11th, Page No. 95)

126. (4)

Asexual reproduction in liverworts takes place by fragmentation of thalli, or by the formation of specialised structures called gemmae (sing. gemma). Gemmae are green, multicellular, asexual buds, which develop in small receptacles called gemma cups located on the thalli (both male and female).

(NCERT Ed. 2025-26, 11th, Page No. 29)

127. (3)

In symmetry, the flower may be actinomorphic (radial symmetry) or zygomorphic (bilateral symmetry). When a flower can be divided into two equal radial halves in any radial plane passing through the centre, it is said to be actinomorphic, e.g., mustard, *datura*, chilli. When it can be divided into two similar halves only in one particular vertical plane, it is zygomorphic, e.g., pea, gulmohur, bean, *Cassia*. A flower is asymmetric (irregular) if it cannot be divided into two similar halves by any vertical plane passing through the centre, as in canna.

(NCERT Ed. 2025-26, 11th, Page No. 62)

[Contd...

128. (4)

Unlike bryophytes and pteridophytes, in gymnosperms, the male and the female gametophytes do not have an independent free-living existence. The multicellular female gametophyte is also retained within megasporangium.

(NCERT Ed. 2025-26, 11th, Page No. 33)

129. (2)

Photosynthetic organisms: Desmids, *Wolffia*

Photosynthetic but partially heterotrophic organisms: *Euglena*, Venus fly trap

Heterotrophic organisms: Yeast, *Neurospora*

Parasitic organisms: *Trypanosoma*, Tobacco mosaic virus *Plasmodium*, *Albugo*, *Cuscuta*, Bacteriophage

(NCERT Ed. 2025-26, 11th, Page No. 17)

130. (2)

The pollen grains and ovules are developed in specialised structures called flowers. *Sequoia* is a gymnosperm.

(NCERT Ed. 2025-26, 11th, Page No. 34)

131. (3)

In palmately compound leaves, the leaflets are attached at a common point, i.e., at the tip of petiole, as in silk cotton.

(NCERT Ed. 2025-26, 11th, Page No. 61)

132. (2)

Guava plants exhibit opposite phyllotaxy. In guava, the flower is epigynous, meaning the thalamus grows upwards and encloses the ovary completely, fusing with it. As a result, the other floral parts like sepals, petals, and stamens are attached above the ovary, making the ovary inferior. Guava fruits typically develop after fertilization and contain numerous seeds and usually not parthenocarpic.

(NCERT Ed. 2025-26, 11th, Page No. 63)

133. (4)

When more than one carpel is present, they may be free (as in lotus and rose) and are called apocarpous. When stamens are attached to the petals, they are epipetalous as in brinjal, or epiphyllous when attached to the perianth as in the flowers of lily. The stamens in a flower may either remain free (polyandrous) or may be united in varying degrees. The stamens may be united into one bunch or one bundle (monoadelphous) as in china rose, or two bundles (diadelphous) as in pea, or into more than two bundles (polyadelphous) as in citrus.

(NCERT Ed. 2025-26, 11th, Page No. 64)

134. (1)

Two types of shoots, long and dwarf, are found in *Ginkgo*.



Ginkgo

(NCERT Ed. 2025-26, 11th, Page No. 33)

135. (4)

In the S phase, DNA content will double (with each chromosome now consisting of two identical sister chromatids) and number of chromosomes will remain same. So the cell will enter the M phase with 4C content and 2n chromosomes. During anaphase, the sister chromatids separate and move to opposite poles. At this point, each chromatid is considered a full chromosome. During telophase and cytokinesis, the cell divides into two. Since the parent cell had 2n chromosomes (each with two chromatids) and these chromatids separated, each daughter cell will receive 2n chromosomes (each with a single chromatid). Thus, the chromosome number is restored to 2n. Since the parent cell had 4C DNA content, and this was equally divided between the two daughter cells, each daughter cell will have a DNA content of 2C.

(NCERT Ed. 2025-26, 11th, Page No. 121)

[Contd...

136. (3)

In a resting neuron, the axonal membrane has many potassium leak channels, so it is much more permeable to K^+.

At the same time, there are very few sodium leak channels, so the membrane is nearly impermeable to Na^+ under resting conditions.

In a resting state, the outer surface of the axonal membrane possesses a positive charge, while the inner surface possesses a negative charge. This electrical gradient (the Resting Membrane Potential) is typically around -70 mV.

(NCERT Ed. 2025-26, 11th, Page No. 232)

137. (4)

Hormone receptors are like a "lock and key." Their molecular shape and chemical properties ensure they only bind to their specific hormone, allowing the endocrine system to send precise signals without interfering with other cellular processes.

Once a hormone binds to its receptor, it triggers a cascade of events. This could include activating enzymes, changing membrane permeability, or stimulating the synthesis of specific proteins, which collectively result in the physiological effect on the target tissue.

Steroid hormones (like estrogen or testosterone) are lipid-soluble and can cross the cell membrane. They bind to intracellular receptors (in the cytoplasm or nucleus). The resulting complex binds to specific DNA sequences to "turn on" or "turn off" specific genes.

Water-soluble hormones (like protein hormones or adrenaline) cannot cross the lipid bilayer of the cell membrane. Instead, they bind to receptors on the cell surface and utilize second messengers (like cAMP or IP3) to carry the signal into the cell.

(NCERT Ed. 2025-26, 11th, Page No. 247)

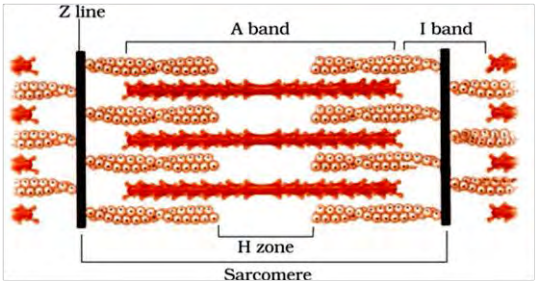
138. (4)

The ascending limb is composed of a thin and a thick segment. Unlike most other parts of the nephron, its walls lack aquaporins (water channels), making it impermeable to water. However, it is highly efficient at moving electrolytes (like Na^+, K^+, and Cl^-) out of the tubule into the medullary interstitium.

The descending limb has a high density of aquaporins, allowing water to move freely out of the tubule and into the salty medullary space via osmosis. Conversely, it has very low permeability to ions like Na^+ and Cl^-.

(NCERT Ed. 2025-26, 11th, Page No. 209)

139. (2)



(NCERT Ed. 2025-26, 11th, Page No. 220)

140. (2)

List-I	List-II
Squamous epithelium	Walls of blood vessels and air sacs
Cuboidal epithelium	PCT of Nephron
Columnar epithelium	Lining of stomach and intestine
Ciliated epithelium	Bronchioles and Fallopian tubes

(NCERT Ed. 2020-21, 11th, Page No. 101)

141. (3)

The sodium-potassium pump is an electrogenic pump. It actively moves 3 Na^+ ions out of the cell (efflux) for every 2 K^+ ions it moves into the cell (influx), contributing to the negative charge inside the membrane (-70mV).

(NCERT Ed. 2025-26, 11th, Page No. 232)

142. (2)

The Peripheral Nervous System (PNS) consists of all the nerves that lie outside the brain and spinal cord. It acts as the communication network connecting the Central Nervous System (CNS) to the limbs and organs.

The nerve fibres of the PNS are divided into two types based on the direction of the impulse:

1. **Afferent (Sensory) fibres:** Carry impulses from tissues/organs to the CNS.
2. **Efferent (Motor) fibres:** Carry regulatory impulses from the CNS to the peripheral tissues/organs.

Homeostasis is the "steady state" or internal balance of the body. Coordination ensures that different systems (like the respiratory and circulatory systems) work together—for example, increasing both heart rate and breathing during exercise to meet oxygen demands.

Neurons are highly specialized, excitable cells.

Their dendrites receive signals, the cell body processes them, and the axon transmits the impulse to the next cell or organ.

(NCERT Ed. 2025-26, 11th, Page No. 231)

[Contd...

143. (2)

Cardiac Output = Heart Rate × Stroke Volume = 80 × 60 = 4800 mL = 4.8 L/min.

Duration of Cardiac Cycle = 60 seconds / Heart Rate = 60 / 80 = 0.75 seconds.

(NCERT Ed. 2025-26, 11th, Page No. 200)

144. (3)

List-I	List-II
Glucagon	Stimulates glycogenolysis
ANF	Decreases blood pressure
Cortisol	Anti-inflammatory reactions
Thyrocalcitonin	Decreases blood calcium levels

(NCERT Ed. 2025-26, 11th, Page No. 245)

145. (2)

Foregut includes pharynx, oesophagus, crop and gizzard.

Foregut and hindgut are ectodermal in origin.

Therefore, both are lined internally by cuticle.

The hindgut is broader than midgut and is differentiated into ileum, colon and rectum.

(NCERT Ed. 2020-21, 11th, Page No. 113)

146. (1)

Muscle contraction is initiated by a signal from the CNS (Central Nervous System) via a motor neuron, not a sensory neuron.

During muscle contraction, the I-band (Isotropic band) shortens.

(NCERT Ed. 2025-26, 11th, Page No. 222)

147. (2)

Fibrous: No movement.

Cartilaginous: Limited movement (e.g., vertebrae, pubic symphysis).

Synovial (Gliding/Pivot): Considerable movement.

(NCERT Ed. 2025-26, 11th, Page No. 227)

148. (2)

Hyposecretion of Vasopressin (ADH) leads to Diabetes Insipidus (water loss), not Diabetes Mellitus (sugar disorder related to Insulin).

(NCERT Ed. 2025-26, 11th, Page No. 242)

149. (1)

FRC = ERV + RV.

FRC = 1000 mL + 1200 mL = 2200 mL.

(NCERT Ed. 2025-26, 11th, Page No. 187)

150. (1)

Usually, the first Rh-positive child is safe because the maternal and foetal blood are well-separated by the placenta. However, during delivery, there is a possibility of the mother's blood being exposed to small amounts of Rh-positive foetal RBCs. This exposure causes the Rh-negative mother to produce Rh antibodies in her blood. If the mother carries another Rh-positive foetus, her immune system is already "primed." The antibodies are already present and ready to attack. These maternal IgG antibodies leak into the foetal circulation. Because they recognize the Rh antigen on the foetal RBCs as foreign, they bind to and destroy the foetal red blood cells (hemolysis). This leads to severe anemia, jaundice, and in extreme cases, can be fatal to the foetus or newborn.

(NCERT Ed. 2025-26, 11th, Page No. 196)

151. (4)

The visceral (autonomic) nervous system is a part of the PNS, includes nerve fibres, ganglia, and plexuses, and has both afferent (sensory) fibres (viscera → CNS) and efferent (motor) fibres (CNS → viscera).

(NCERT Ed. 2025-26, 11th, Page No. 231)

152. (2)

The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner so as to leave some minute spaces called filtration slits or slit pores.

(NCERT Ed. 2025-26, 11th, Page No. 208)

153. (1)

The adrenal medulla secretes Adrenaline (Epinephrine) and Noradrenaline (Norepinephrine). These are collectively called catecholamines or "Fight or Flight" hormones. They increase alertness, pupil dilation, heart rate, and the rate of respiration during emergency situations.

The adrenal cortex is divided into three layers, and the Zona glomerulosa is actually the outermost layer. The correct order from outside to inside is:

1. Zona Glomerulosa (Outer - secretes mineralocorticoids)
2. Zona Fasciculata (Middle - secretes glucocorticoids)
3. Zona Reticularis (Inner - secretes androgens)

Cortisol (the main glucocorticoid) helps maintain blood pressure, cardio-vascular functions and as well as the kidney functions.

Aldosterone stimulates the reabsorption (saving) of Na^+ and water from the renal tubules, while stimulating the excretion of K^+ and phosphate ions. Its primary goal is to increase blood volume and pressure.

(NCERT Ed. 2025-26, 11th, Page No. 245)

[Contd...

154. (3)

In males cockroach, genital pouch or chamber lies at the hind end of abdomen bounded dorsally by 9th and 10th terga and ventrally by the 9th sternum.

(NCERT Ed. 2020-21, 11th, Page No. 112)

155. (4)

Inside the skull, the brain is covered by cranial meninges consisting of an outer layer called dura mater, a very thin middle layer called arachnoid and an inner layer (which is in contact with the brain tissue) called pia mater.

(NCERT Ed. 2025-26, 11th, Page No. 235)

156. (2)

A-band represents the length of thick (myosin) filaments. Since myosin filaments do not shorten, the A-band remains constant during contraction.

I-band = region with only actin filaments

As actin slides inward, the I-band becomes narrower.

H-zone = region with only myosin

With actin sliding inward, overlap increases → H-zone reduces or may disappear.

Z-lines are attached to actin.

Sliding pulls actin toward the center, so Z-lines move closer, sarcomere shortens.

(NCERT Ed. 2025-26, 11th, Page No. 222)

157. (3)

Secretin acts on the exocrine pancreas to release water and bicarbonates. CCK stimulates the secretion of pancreatic enzymes and bile.

(NCERT Ed. 2025-26, 11th, Page No. 247)

158. (2)

The respiratory rhythm centre is in the medulla and is responsible for generating the basic rhythm of breathing. The pneumotaxic centre, located in the pons, can moderate/regulate the activity of the respiratory rhythm centre.

Pneumotaxic centre controls the duration of inspiration mainly by sending signals that reduce the duration of inspiration. When inspiration duration is reduced, breathing rate increases.

(NCERT Ed. 2025-26, 11th, Page No. 190)

159. (1)

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	Nil	AB, A, B, O
O	Nil	anti-A, B	O

(NCERT Ed. 2025-26, 11th, Page No. 195)

160. (1)

Insulin is a hypoglycemic hormone because it lowers blood glucose level.

Insulin:

- acts on hepatocytes (liver cells) and adipocytes (fat cells) enhances uptake and utilisation of glucose stimulates glycogenesis (conversion of glucose to glycogen in liver and muscles)

(NCERT Ed. 2025-26, 11th, Page No. 245)

161. (2)

The Limbic system is involved in regulation of sexual behaviour, expression of emotional reactions (excitement, pleasure, rage, fear), and motivation.

(NCERT Ed. 2025-26, 11th, Page No. 236)

162. (1)

In the inner medullary collecting duct, urea can diffuse out into the medullary interstitium. This helps maintain high osmolarity in the medulla, which is essential for concentrating urine.

The lungs are considered accessory excretory organs.

- CO_2 Removal: On average, a resting human body eliminates approximately 200 mL of CO_2 per minute.
- Water Loss: We also lose water in the form of water vapor through expired air (insensible water loss). While the exact amount varies based on humidity and activity levels, it is considered a "significant" part of our daily water output.

(NCERT Ed. 2025-26, 11th, Page No. 213)

163. (3)

Blood is a specialised connective tissue containing a fluid matrix (plasma) and formed elements. It typically lacks fibres (fibrinogen is present but fibres form only during clotting). Cartilage and bone have solid/pliable matrices with fibres.

(NCERT Ed. 2020-21, 11th, Page No. 104)

164. (4)

List-I	List-II
Thalamus	Major coordinating centre for sensory and motor signaling
Cerebellum	Balance and convoluted surface for neurons
Medulla oblongata	Cardiovascular reflexes and gastric secretions
Hypothalamus	Body temperature, urge for eating

(NCERT Ed. 2025-26, 11th, Page No. 236)

165. (2)

The Anterior Pituitary (Adenohypophysis) is regulated by hormones from the hypothalamus reaching it via the Hypophyseal Portal System. The Posterior Pituitary (Neurohypophysis) is under direct neural regulation (axons of hypothalamic neurons end there).

(NCERT Ed. 2025-26, 11th, Page No. 240)

166. (4)

The pectoral girdle consists of a clavicle and a scapula. The scapula has a depression called the glenoid cavity which articulates with the head of the humerus to form the shoulder joint.

(NCERT Ed. 2025-26, 11th, Page No. 226)

167. (4)

Clotting occurs through a series/cascade of reactions involving clotting factors.

Thrombin acts on soluble fibrinogen and converts it into insoluble fibrin, forming the clot network.

Thrombokinase (prothrombin activator) helps convert prothrombin → thrombin.

Ca^{2+} ions are essential cofactors at multiple steps in the clotting pathway.

(NCERT Ed. 2025-26, 11th, Page No. 196)

168. (4)

Vertebrates have a more developed nervous system, especially the central nervous system (brain and spinal cord).

Invertebrates generally have ganglia and nerve cords, which are less complex compared to the vertebrate brain.

Synapse is the junction between two neurons, involving the pre-synaptic and post-synaptic membranes (with synaptic cleft in between).

Hypothalamus is part of the forebrain (diencephalon) and regulates homeostasis, temperature, hunger, thirst, and controls the endocrine system through the pituitary.

The electrical potential difference across the resting neural membrane is called the 'resting potential'.

(NCERT Ed. 2025-26, 11th, Page No. 231)

169. (2)

Melatonin is secreted by the Pineal gland and regulates circadian rhythms.

(NCERT Ed. 2025-26, 11th, Page No. 242)

170. (3)

Asthma = Inflammation of bronchi/bronchioles.

Emphysema = Alveolar wall damage (Cigarette smoking).

Silicosis/Asbestosis = Occupational disorders leading to fibrosis.

(NCERT Ed. 2025-26, 11th, Page No. 191)

171. (1)

List-I	List-II
PCT	Reabsorption of 70-80% electrolytes and nutrients
Loop of Henle	Concentration of urine (Counter current)
DCT	Conditional reabsorption of Na^+ and water
Bowman's capsule	Ultrafiltration

(NCERT Ed. 2025-26, 11th, Page No. 209)

172. (3)

Female cockroaches possess a pair of spermathecae in the 6th abdominal segment which opens into the genital chamber to store sperm.

Anal styles are a key feature of sexual dimorphism in cockroaches. They are found on the 9th sternite of males only; females lack them. (Note: Both sexes have anal cerci on the 10th segment).

(NCERT Ed. 2020-21, 11th, Page No. 112)

173. (1)

GIP stands for Gastric Inhibitory Peptide. As the name suggests, it inhibits gastric secretion and motility.

(NCERT Ed. 2025-26, 11th, Page No. 247)

174. (2)

The Nodes of Ranvier are the unmyelinated gaps where ion exchange occurs during saltatory conduction.

(NCERT Ed. 2025-26, 11th, Page No. 232)

175. (2)

Red muscle fibres are named so because of their high myoglobin (oxygen-binding pigment) content. While they have many mitochondria, they actually have a relatively low amount of sarcoplasmic reticulum compared to white muscle fibres. High sarcoplasmic reticulum is a feature of white fibres (specialized for fast contraction). Because they have high myoglobin and many mitochondria, they can utilize oxygen efficiently and depend on aerobic respiration.

These fibres are "slow-twitch" and do not fatigue quickly; they are specialized for sustained work (e.g., postural muscles). White fibres, which depend on anaerobic respiration, are the ones that fatigue quickly.

(NCERT Ed. 2025-26, 11th, Page No. 223)

176. (4)

Lymph is a colorless fluid that contains specialized lymphocytes (a type of White Blood Cell) which are responsible for the immune responses of the body.

Lymph (as interstitial fluid) acts as an important carrier for nutrients, hormones, and gases, facilitating exchange between the blood capillaries and the tissue cells.

In the intestinal villi, specialized lymphatic capillaries called lacteals are responsible for the absorption of fats (chylomicrons).

While lymph has the same mineral distribution as plasma, it is characterized by the absence of RBCs (Red Blood Cells) and a lower concentration of proteins. It does not possess more RBCs than blood; it possesses none.

(NCERT Ed. 2025-26, 11th, Page No. 197)

177. (2)

Protein/peptide hormones are water-soluble, so they cannot enter the cell.

They bind to membrane-bound receptors and trigger formation of second messengers inside the cell to bring about the response.

Common second messengers such are:

- cAMP
- Ca^{2+}
- IP_3 (Inositol triphosphate)

(NCERT Ed. 2025-26, 11th, Page No. 248)

178. (2)

Statement I is correct. Statement II is incorrect because saltatory conduction is faster (not slower) than continuous conduction in non-myelinated fibres.

(NCERT Ed. 2025-26, 11th, Page No. 232)

179. (3)

A fall in Glomerular Filtration Rate (GFR) is the primary trigger for the RAAS pathway. JG cells act as baroreceptors/chemoreceptors that detect a drop in blood pressure or volume and respond by secreting the enzyme renin.

Angiotensin II is a powerful vasoconstrictor, not a vasodilator. Its function is to constrict blood vessels to increase blood pressure and restore GFR.

(NCERT Ed. 2025-26, 11th, Page No. 212)

180. (1)

Alveoli pCO_2 = 40 mm Hg.

Deoxygenated blood pCO_2 = 45 mm Hg.

(Diffusion occurs from blood to alveoli).

(NCERT Ed. 2025-26, 11th, Page No. 187)

■■■



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if

[Contd...